Chameleon Solutions with Physics-Informed Neural Networks

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Dark Energy & The Fifth Force Problem

A light scalar field is a compelling dark-energy candidate. If coupled to matter, it mediates a fifth force constrained by precision-gravity tests. A viable scalar field must therefore screen itself in dense environments while remaining active in the cosmological vacuum.

The Chameleon Mechanism

Proposed by Khoury & Weltman (2004), the chameleon acquires a **density-dependent effective mass** through competition between a self-interaction potential and a matter coupling:

$$V_{\rm eff}(\phi) = \frac{\Lambda^{4+n}}{\phi^n} + \rho\, e^{\alpha\phi} \tag{1}$$

The minimum and curvature of $V_{\rm eff}$ shift with local density:

$$m^2_{\rm eff}(\rho) = V''_{\rm eff}(\phi_{\rm min}) \propto \rho^{\frac{n+2}{n+1}} \tag{2}$$

In dense matter: large $m_{\rm eff}$, short range $\Rightarrow$ **screened**.
In vacuum: small $m_{\rm eff}$, long range $\Rightarrow$ **observable fifth force**.

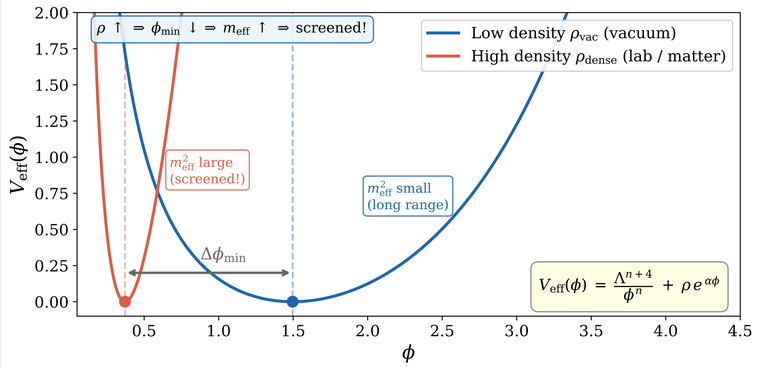


Figure 1: Effective potential $V_{\rm eff}(\phi)$ at low & high density (blue & red). The minimum shifts & the well steepens as the density increases, raising the mass & screening fifth force.

Laboratory probes include atom interferometry, torsion balances, ultracold neutrons and levitated microspheres.

This Work

We use physics-informed neural networks to solve the nonlinear chameleon field equation across physical parameters and full 3D geometries. Solutions are validated against independent radial benchmarks, where available.

PINNs allow us to move from individual field solutions to fast, continuous surrogate models.

References

- Khoury & Weltman, *PRL* **93**, 171104 (2004)
- Lu, Meng, Mao & Karniadakis, *SIAM Rev.* **63**, 208 (2021) [DeepXDE]
- Briddon, Burrage, Moss & Tamošiūnas, *JCAP* **12**, 043 (2021) [SELCIE]

Physics-Informed Neural Networks

A PINN encodes the PDE directly into the training loss. The field $\hat{\phi} = \mathcal{N}(\mathbf{x};\theta)$ is evaluated at collocation points sampled throughout the domain Ω and its boundary $\partial\Omega$:

$$\mathcal{L}(\theta) = \underbrace{\left\|\nabla^2\hat{\phi} - V'(\hat{\phi}) - \rho\, e^{\alpha\hat{\phi}}\right\|^2_{\Omega}}_{\mathcal{L}_{\rm PDE}} + \underbrace{\left\|\hat{\phi} - \phi_{\rm BC}\right\|^2_{\partial\Omega}}_{\mathcal{L}_{\rm BC}} \tag{3}$$

The network parameters θ are optimised using Adam followed by L-BFGS. Spatial derivatives are evaluated by **automatic differentiation**, without finite-difference stencils or an explicit mesh.

Key advantage: the same physics-informed framework extends from radial solutions to full 3D, while the sampling and geometric representation are adapted to the problem.

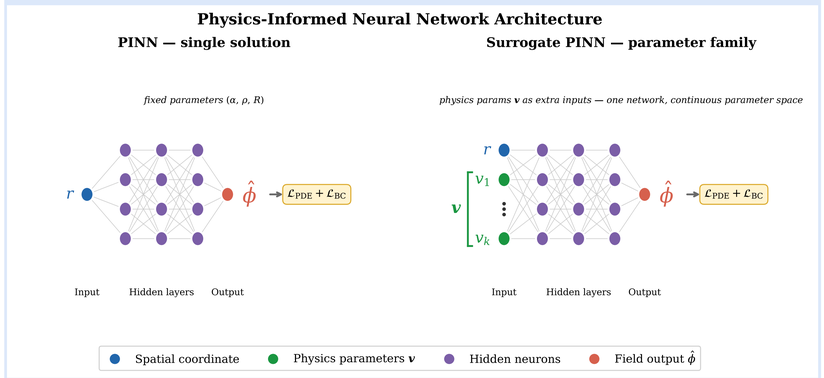


Figure 2: Left: standard PINN with spatial input only. Right: surrogate PINN; physics parameters $\boldsymbol{\nu}$ added as inputs, yielding a continuous family of solutions from a single network.

Surrogate Model: Radial Chameleon

For a dense sphere of radius $R_{\rm obj}$ and density $\rho_{\rm obj}$ in spherical symmetry:

$$\phi''(r) + \frac{2}{r}\phi'(r) = V'(\phi) + \rho(r)\, e^{\alpha\phi} \tag{4}$$

Promoting the **physical parameters** $\boldsymbol{\nu} = (\alpha, \rho_{\rm obj}, R_{\rm obj})$ to **network inputs** yields a continuous **surrogate**: $\hat{\phi} = \mathcal{N}(r,\ \boldsymbol{\nu};\ \theta)$

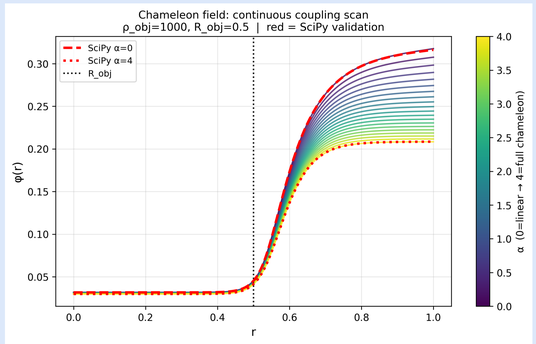


Figure 3: Chameleon field $\phi(r)$ across $\alpha \in [0, 4]$ at fixed $\rho_{\rm obj} = 1000$ and $R_{\rm obj} = 0.5$. Dashed red curves show independent validation solutions at $\alpha = 0$ and $\alpha = 4$.

Once trained, surrogate enables rapid parameter-space exploration.

Full 3D: Single Sphere

Moving to a Cartesian domain $[-L, L]^3$, we solve the 3D chameleon equation with **no imposed symmetry**:

$$\nabla^2\phi = V'(\phi) + \rho(\mathbf{x})\, e^{\alpha\phi} \tag{5}$$

For both smoothly regularised and sharp density profiles at the sphere boundary, the PINN recovers the screened interior & an approximately spherical field profile.

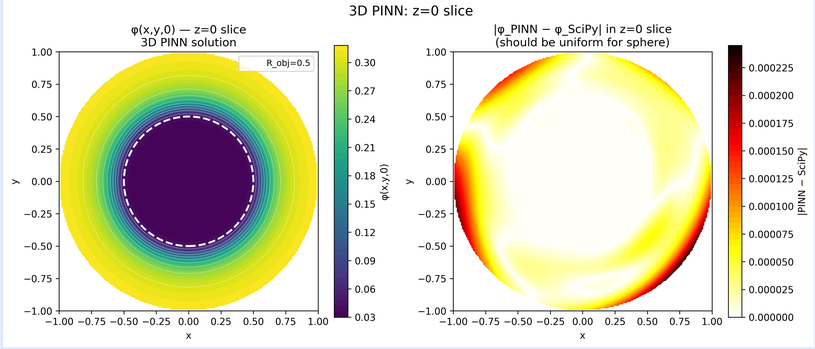


Figure 4: Smooth-boundary benchmark for a dense sphere ($R = 0.5$). Left: PINN solution $\hat{\phi}(x, y, 0)$. Right: pointwise error relative to an independent radial reference.

3D Two-Sphere Surrogate: Collective Screening

Taking two spheres with the centre-to-centre separation d as a network input,

$$\hat{\phi} = \mathcal{N}(x, y, z,\ d\ ;\ \theta)\,, \tag{6}$$

a single surrogate represents a continuous family of two-sphere solutions.
Three screening regimes emerge:

- **Merged** (d small): midpoint field $\phi(0,0,0) \approx \phi_{\rm obj}$ — fully screened
- **Transition**: rapid rise as spheres separate and screening weakens
- **Independent** (d large): $\phi(0,0,0) \to \phi_{\rm vac}$ — objects decouple

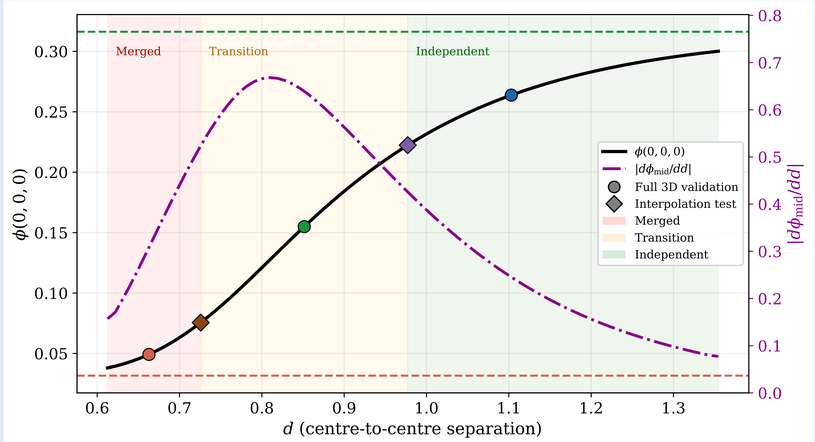


Figure 5: Midpoint field $\phi(0,0,0)$ as a function of separation d. The surrogate prediction is compared with independent full-3D solutions at selected separations.

New separations are evaluated directly—no re-solving required.